

Research project at SNS

From february 2025 to july 2025

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SCUOLA
NORMALE
SUPERIORE

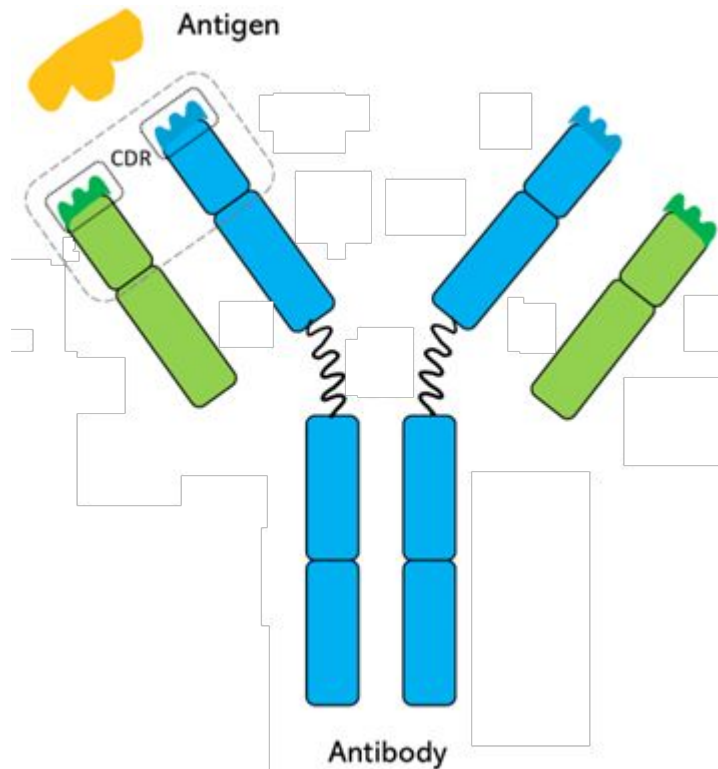


Antibody-Antigen affinity prediction from amino acid sequences

Summary

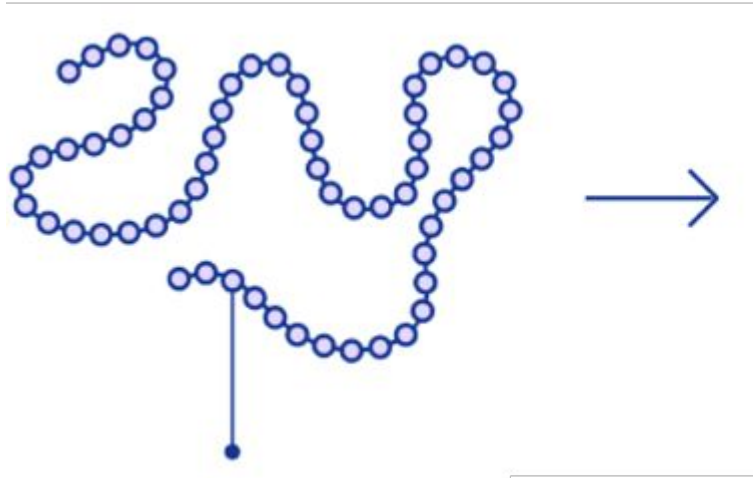
Introduction and motivations

- I. First prediction models
- II. Data Augmentation
- III. Augmented models



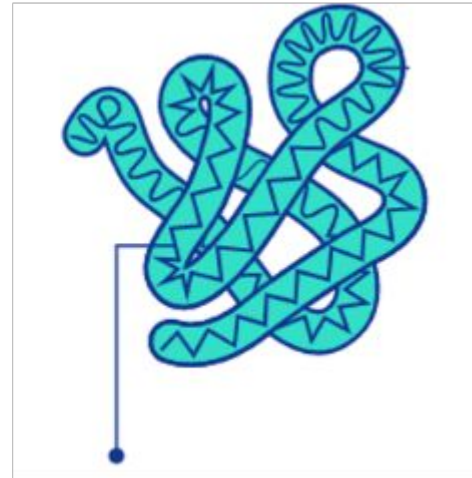
Introduction - from sequence to structure

A complex biological process



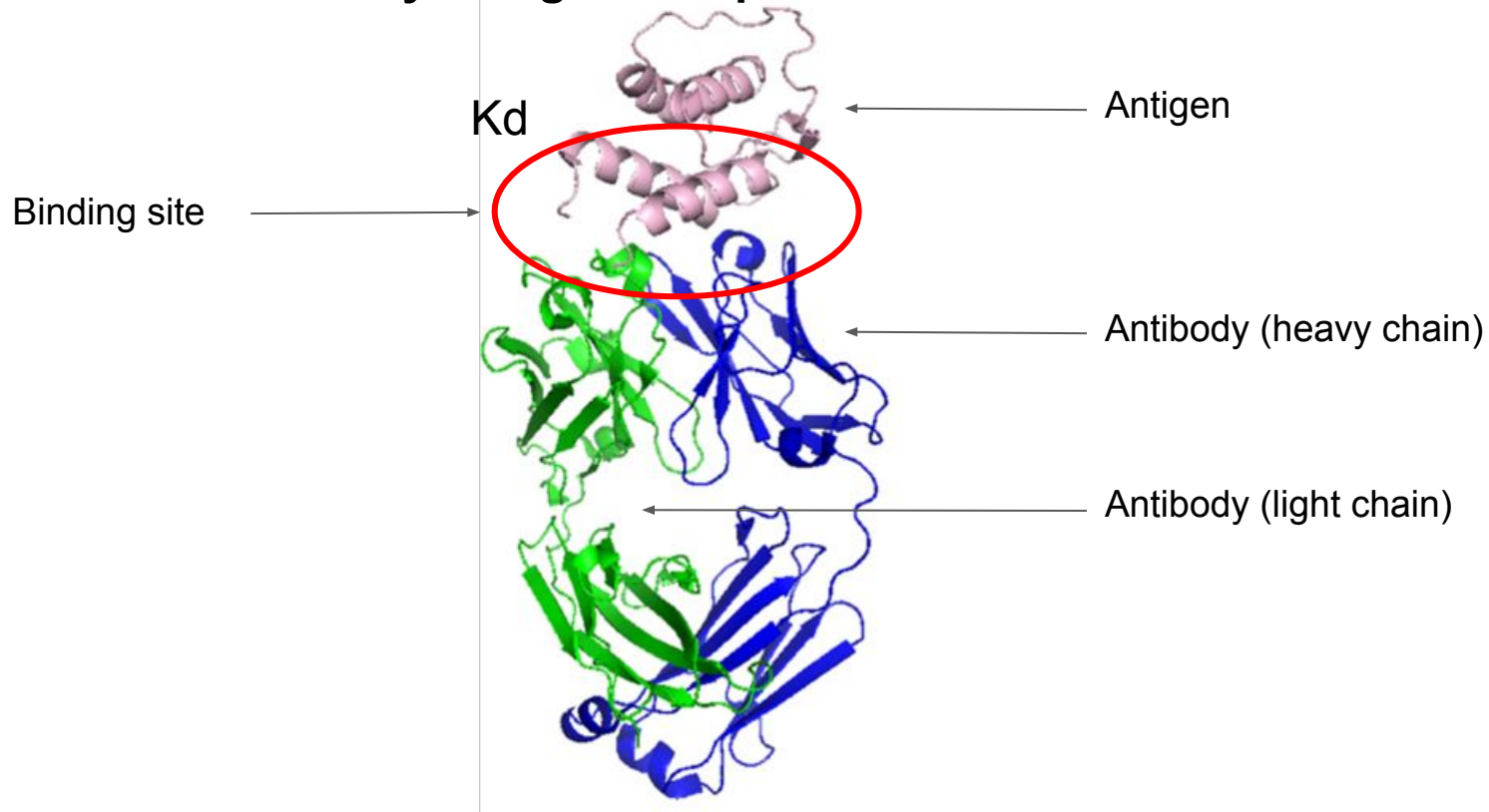
amino acid sequence

SITLTQSPDSATF GTSDVTFGQGTTVNTTTPP



3D structure

Introduction - Antibody-Antigen complexes



Dissociation constant K_d quantifies the strength of the interaction between the antibody and the antigen

Introduction - Goal of the project

Predict the K_d from amino acid sequences

VTLVQSGDVVAK LSALGYWGQGTLTVVSEA

SITLTQSPDSATF GTSDVTFGQGTTVNTTTPP

KTLEDVVNDVK ELSQVDELSKTPDPNVNFN



K_d

Introduction - Data Base

SabDab database

Heavy chain

Light chain

Antigen

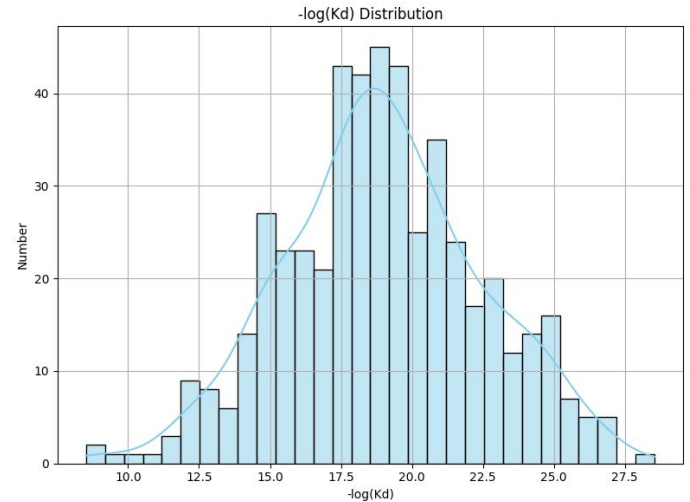
SITLTQSPDSATF GTSDVTFGQGTTVNTTTPP

VTLVQSGDVVAK LSALGYWGQGLTVTSEA

KTLEDVVNDVK ELSQVDELSKTPDPNVNFN

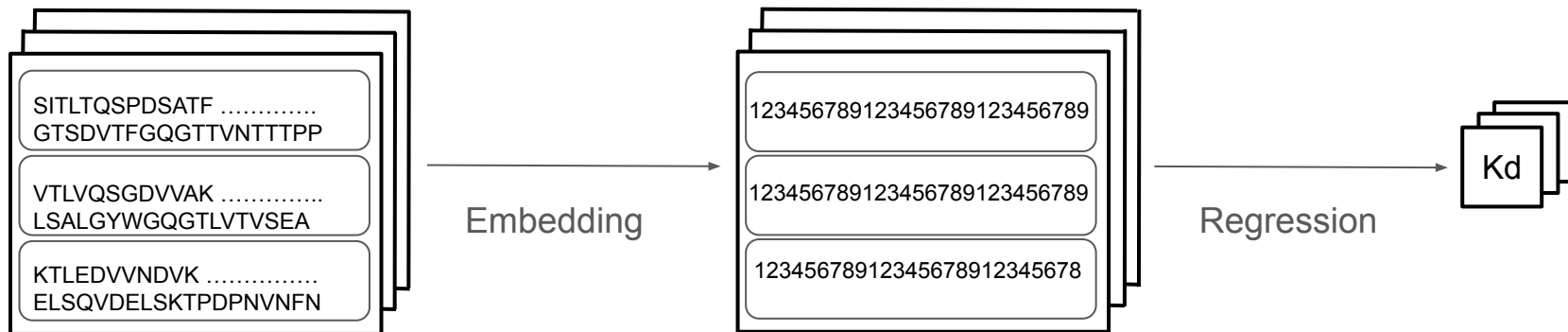
~ 400 complexes

K_d



Introduction - Methods

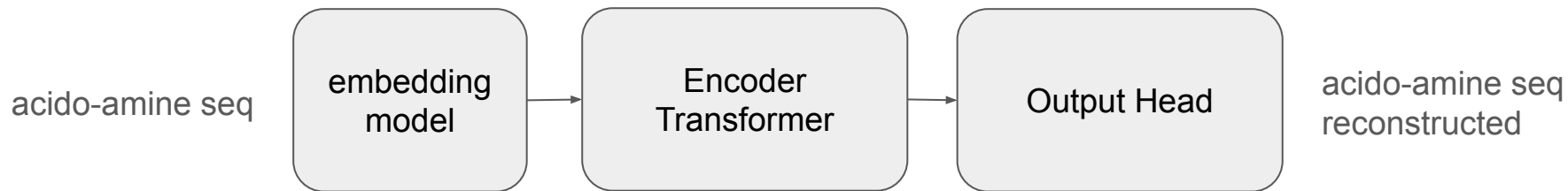
Pipeline



I. Prediction Models

1. Embedding models

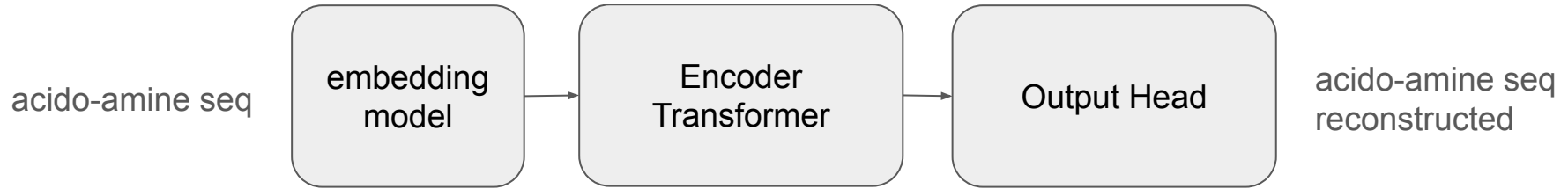
Protein Language Models : autoregressive models trained on proteins



I. Prediction Models

1. Embedding models

Protein Language Models : autoregressive models trained on proteins



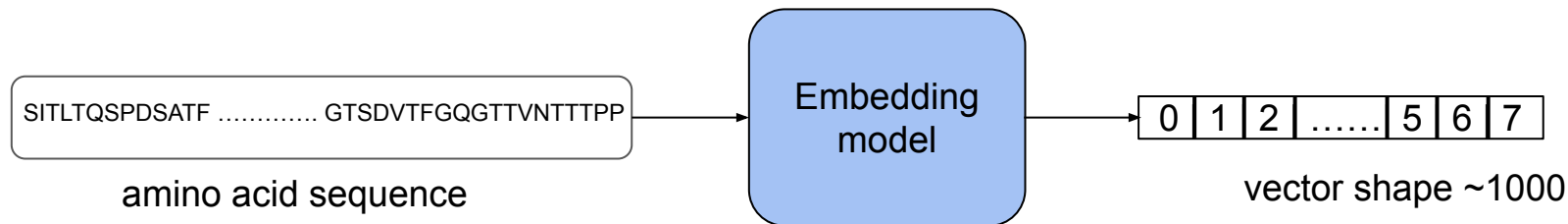
We are focusing on a specific subset of proteins : antigens / antibodies. We aim to downsize the embedding model and retrain it on our subset

We used Protbert, Boltz-2, ESM2 models

I. Prediction Models

1. Embedding models

Embedding : downsize and finetuning



Method 1 :

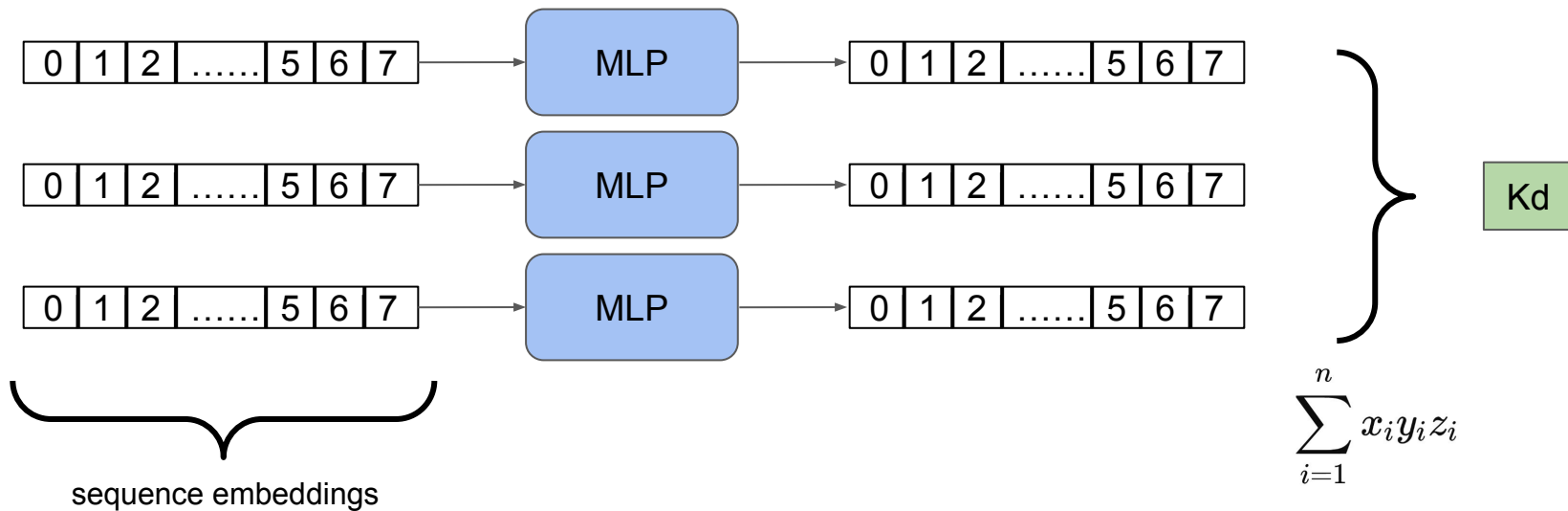
- Downsize with PCA space W (dim 128)
- Domain adaptation by finetuning on our antibody / antigen subset with $\nabla'_\theta = WW^T \nabla_\theta$

Method 2 :

- Downsize with PCA space W
- add PCA / inverse PCA layer to compute the gradient on our antibody antigen subset

I. Prediction Models

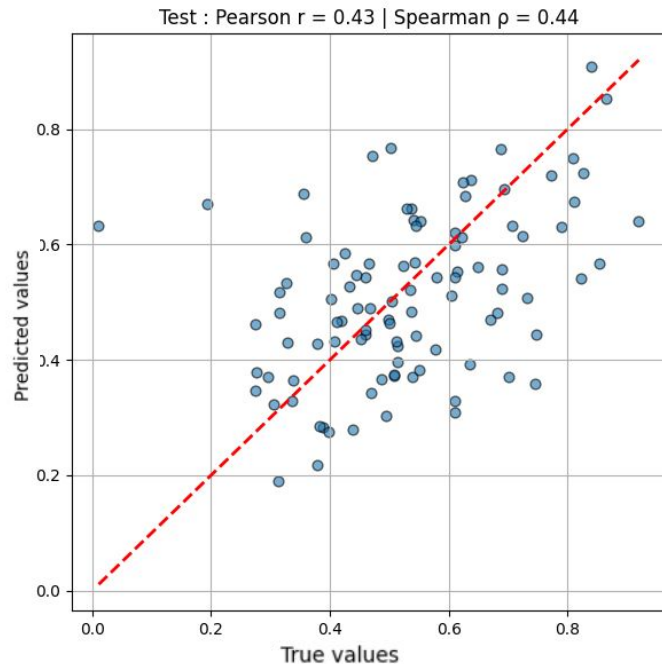
2. Tri-MLP model



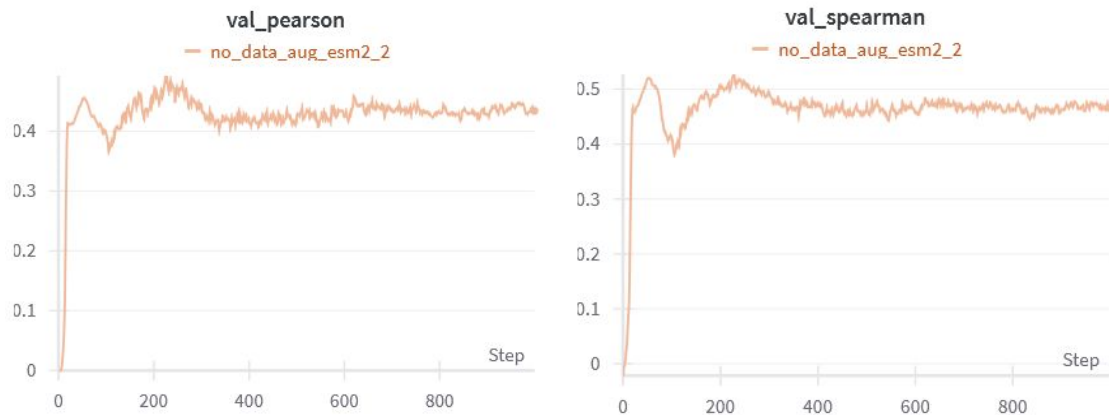
I. Prediction Models

3. First Results

Correlation plot : GT vs Prediction



Correlation scores evolution



correlation scores ~ 0.45

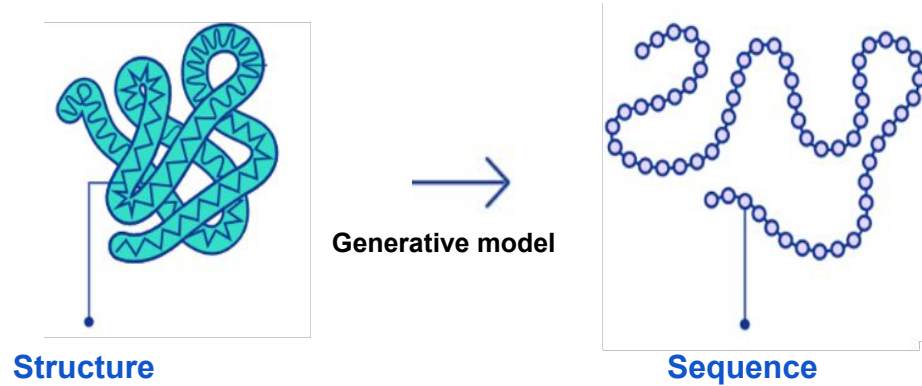
II. **Data augmentation**

Build a new dataset with synthetic data

II. Data augmentation

1. Generative models

Generative models can **predict a protein sequence** that would fold into a **given structure**

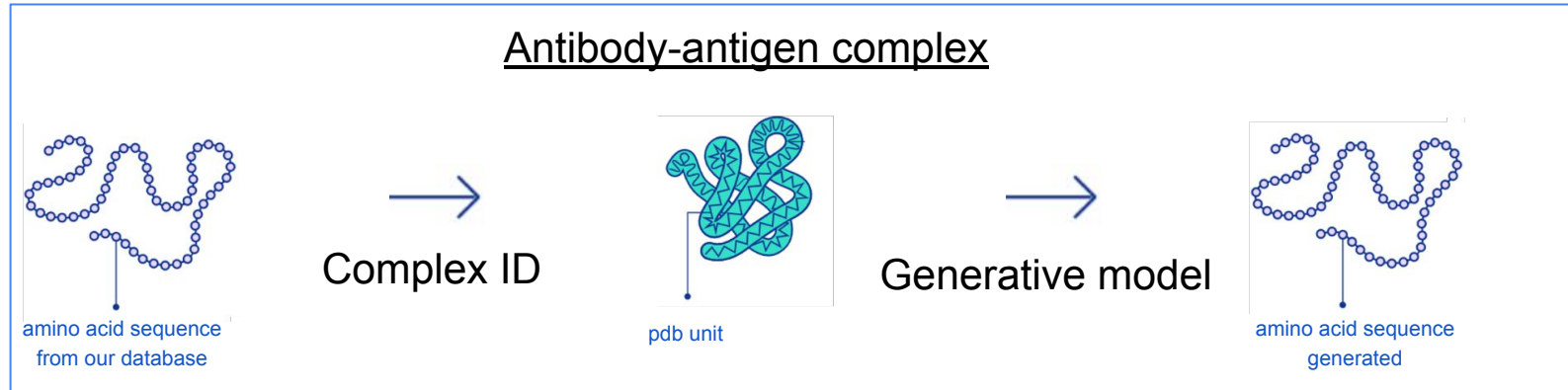


Generative models : ProtMPNN, Boltz-2, Alpha-Fold

II. Data augmentation

2. Create synthetic data to feed the model

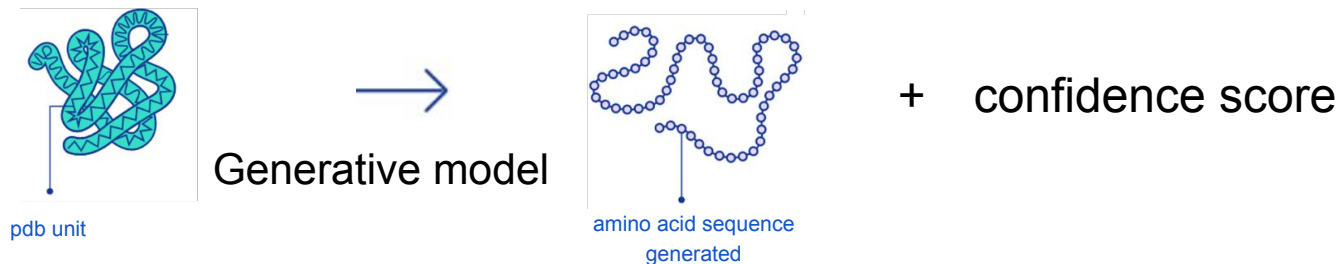
Idea : using generative models such as ProtMPNN / Boltz 2, we can create new sequences



II. Data augmentation

3. Generation step

Generative models give a confidence score while generate sequences



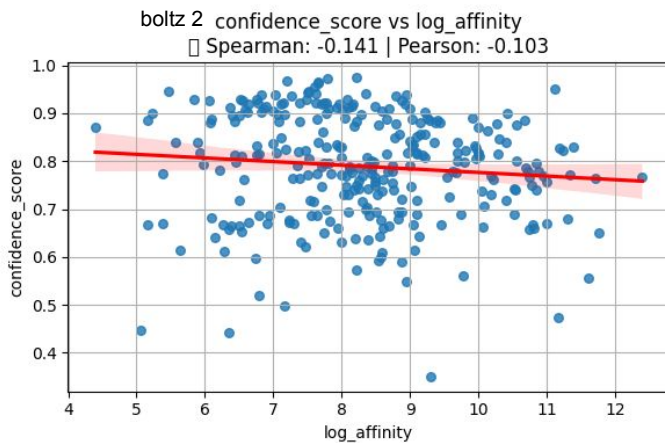
We want to use this score as the **target** for our regression model. A **strong binding affinity** between the antibody and antigen suggests that the generative model should more easily produce high-quality sequences, leading to a **higher confidence score**

Low K_d > High confidence score

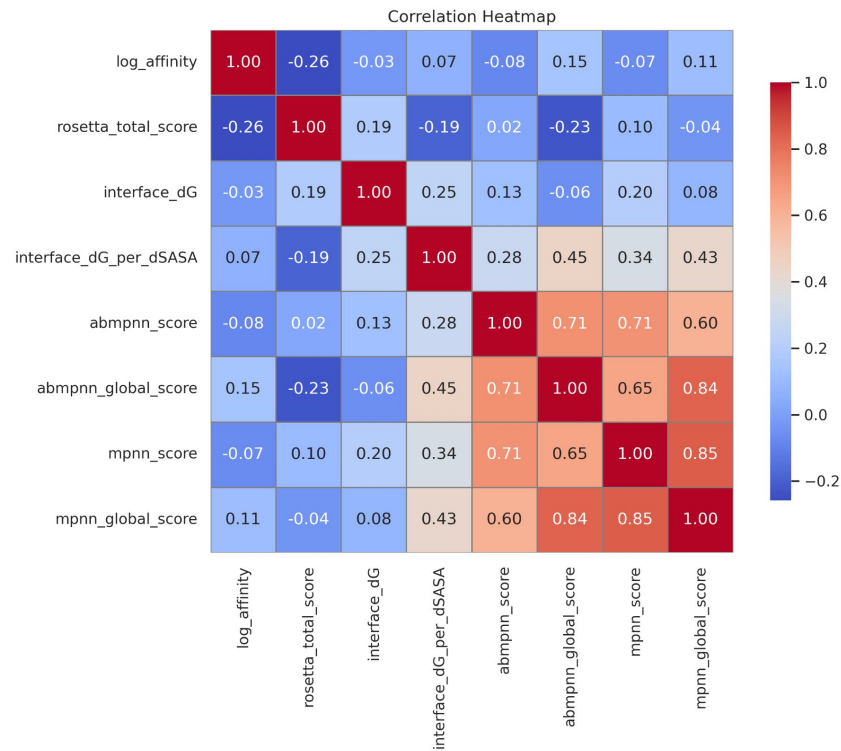
High K_d > Low confidence score

II. Data augmentation

4. What about the correlation Kd (log affinity) / confidence score ?



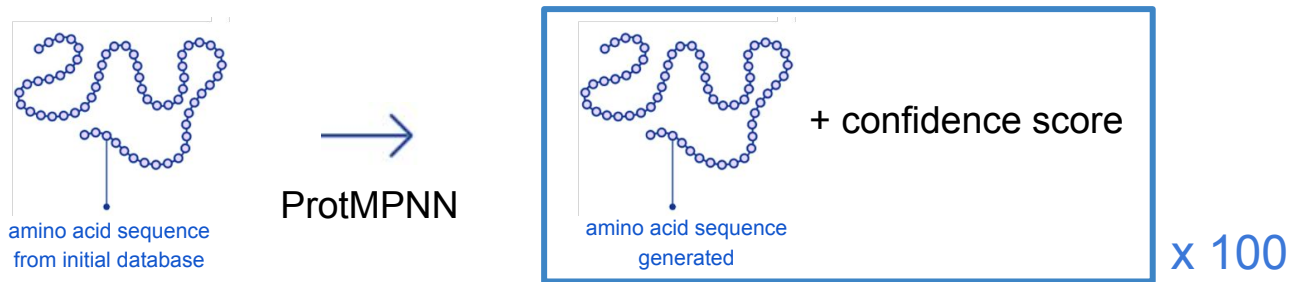
- protMPNN
- finetuned protMPNN
- rosetta
- boltz 2



II. Data augmentation

5. New Dataset

Generation of the new dataset



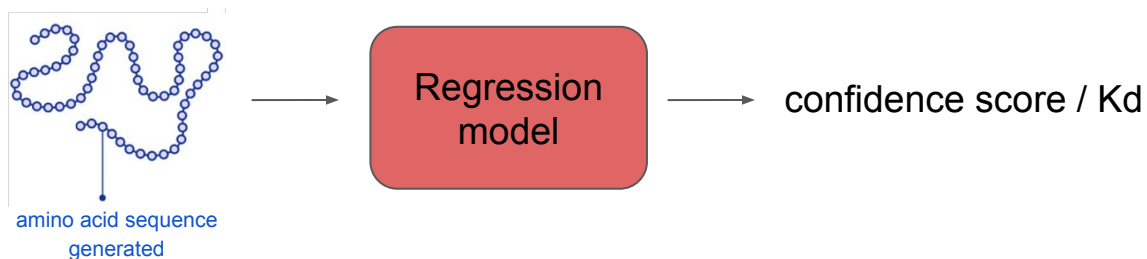
~ 40000 new complexes + confidence scores

~ 400 Kd values

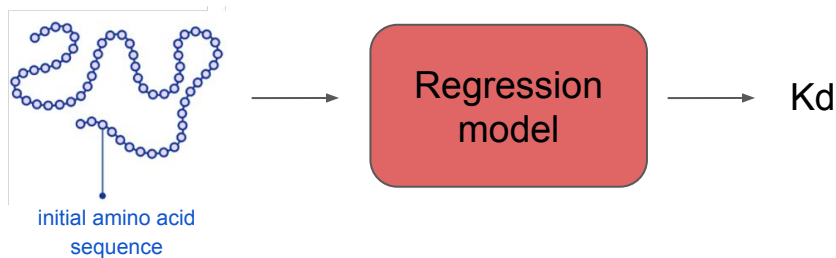
III. Augmented models

Train a model in 2 steps

1. pretraining the model on the new database



2. finetuning the model on the initial database

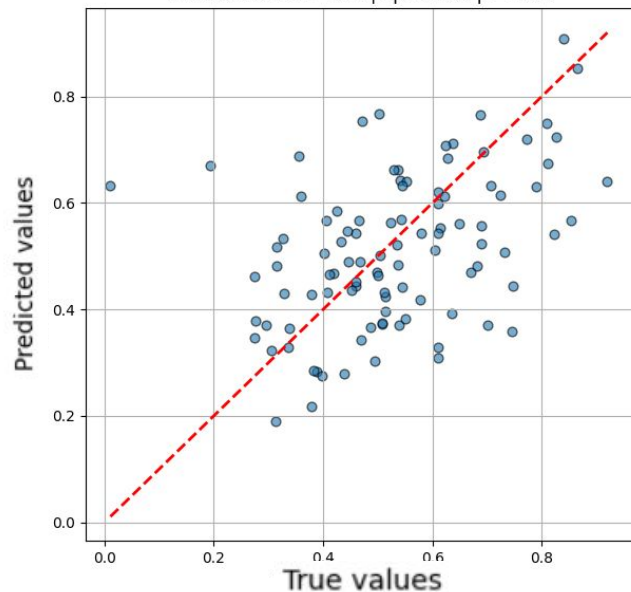


III. Augmented models

Case 1 : Finetuning the model pretrained on confidence score

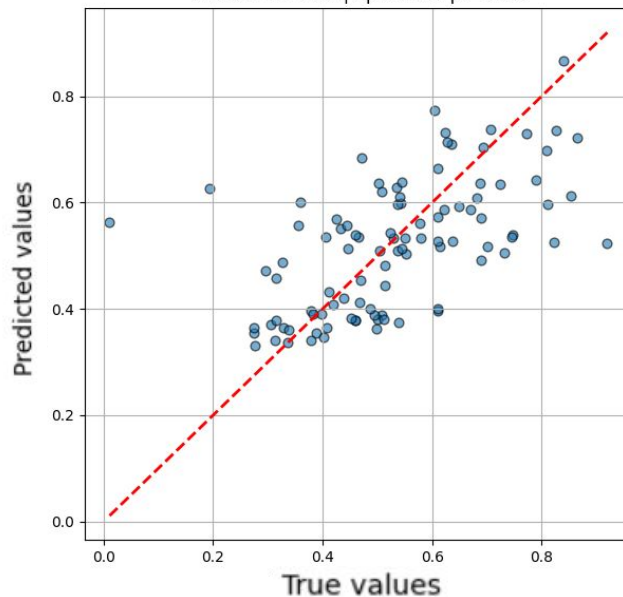
not augmented model

Test : Pearson $r = 0.43$ | Spearman $\rho = 0.44$



augmented model

Pearson $r = 0.58$ | Spearman $\rho = 0.62$

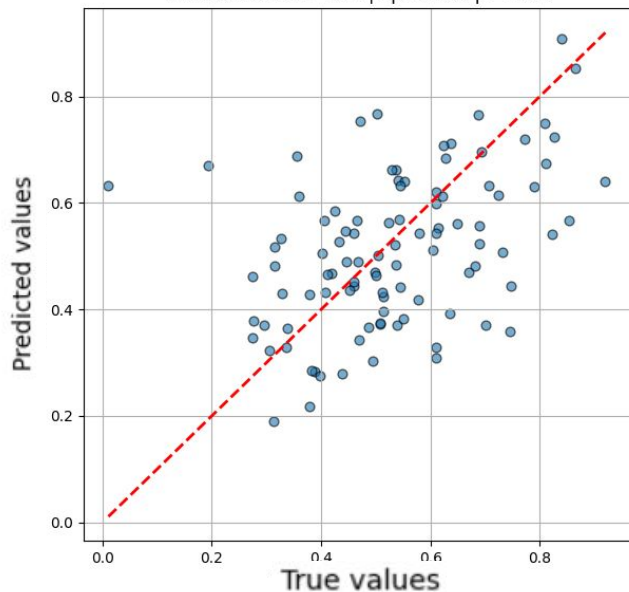


III. Augmented models

Case 2 : Finetuning the model pretrained on Kd

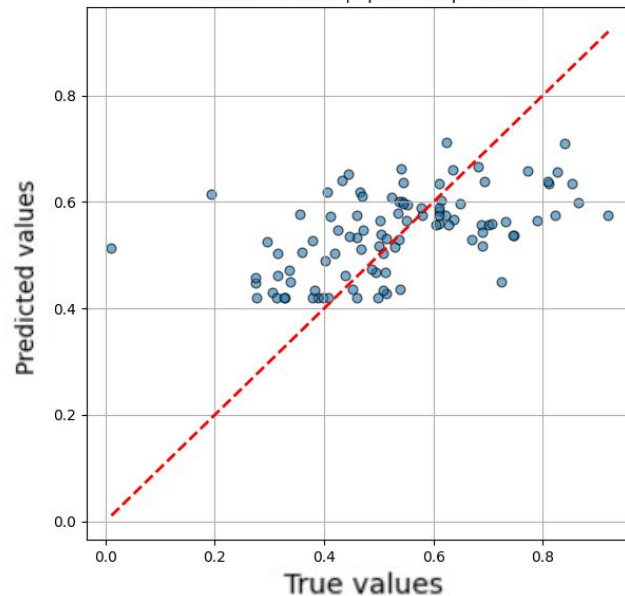
not augmented model

Test : Pearson $r = 0.43$ | Spearman $\rho = 0.44$



augmented model

Pearson $r = 0.53$ | Spearman $\rho = 0.56$



Conclusion

- Very challenging task with limited data
- Significant improvement with data augmentation
- Models still not very performant

To go further :

- Design a more robust model and fine-tune it on the augmented dataset
- Try data augmentation from another model and/or use a different score as the regression target

Thank you all !